# KAILANI ALARCON

**Electrical Engineer | Robotics, Embedded Systems & Verification**

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## EDUCATION

**University of Nevada, Reno | Reno, NV** GPA: 3.2 / 4.0

M.S. Electrical Engineering | Expected May 2028

B.S. Electrical Engineering, Minor in Mathematics | 2026

**Focus:** Robotics, Autonomous Vehicles, and Embedded Systems (RAVES) | Honors Baccalaureate

**Coursework:** Control Systems I & II, Robotics, Embedded Systems, Digital Signal Processing, Microprocessor Applications

## TECHNICAL SKILLS

**Systems & Analysis:** Requirements development, verification planning, risk assessment, BOM development

**Embedded & Controls:** ROS, STM32, UART, SPI, I2C, PWM, interrupts, encoder capture, sensor integration, PID

**Programming & Tools:** C, C++, Python, MATLAB, Simulink, Creo, Thermal Desktop, Arduino, Git, STM32CubeIDE

**Hardware:** Oscilloscopes, DMMs, FPGA/VHDL, VNA/S-parameters, SOLT calibration, FFT, thermal/photometric validation

## EXPERIENCE

**OSTEM Intern, Human Factors Engineering and Mechanical Design | NASA** Aug. 2026–Present

**Marshall Space Flight Center | Huntsville, AL** 40 hrs. per week

- Evaluated a legacy robotic arm for lunar-dust testing in a high-vacuum chamber targeting ~$1 \times 10^{-7}$ Torr, using Python and a Linux development environment to characterize geometry, mass, actuator travel, workspace, and vacuum constraints.
- Converted 6 operational scenarios into 34 hierarchical system requirements covering manipulation, workspace, emergency shutdown, vacuum operation, and verification, incorporating ROS 2 and MuJoCo into robotic control and simulation
- Identified a potential shoulder-actuator torque shortfall, comparing a calculated 27.23 lbf-in shoulder load against estimated actuator capability of 18.6–24.9 lbf-in using Creo and motor data.

**Product Development Intern** Sept. 2023 - Aug. 2026

**LUX Dynamics | Reno, NV** 40 hrs. per week

- Provided electrical, thermal, and photometric validation for 12+ LED products, generating performance data used to support manufacturing-release and engineering decisions.
- Managed 4–5 concurrent product-development projects and presented technical findings weekly to managers, customers, and engineers, supporting coordinated design and validation decisions.
- Improved thermal testing by developing an adopted validation procedure and Python/Arduino automation for acquisition, logging, ambient control, analysis, and reporting; documented results in 3 technical white papers.

**Peer Tutor** Aug. 2024 - May 2026

**Engineering Tutoring Center | University of Nevada, Reno** 20 hrs. per week

- Supported 12–15 engineering students weekly across embedded C, microcontrollers, circuits, and real-time control by providing individualized technical instruction and problem-solving guidance.

## SELECTED PROJECTS

**IVerify Capstone Project | Project and Systems Integration Lead** Spring 2026

- Led systems integration for a 4-person team developing a 5-subsystem non-invasive IV occlusion detector with 3-state classification, coordinating mechanical sensing, analog electronics, firmware, user indication, and verification planning.

**Thermal Controller and Logger | Python, Arduino, TC-08** 2025 - 2026

- Developed a unified thermal-test platform integrating TC-08 data logging, ambient control, live visualization, stability tracking, metadata capture, and automated Excel reporting to streamline test execution and analysis.

**Fingerprint-Based Authentication | STM32F767ZI, UART** Fall 2025

- Built and validated an STM32 biometric authentication system supporting 24 fingerprint templates across 4 users and 2 UART channels operating at 57,600 and 115,200 baud.

**Virtual Network Analyzer and SFCW | MATLAB, SOLT, S-Parameters** Spring 2025

- Developed MATLAB control and 12-term SOLT calibration for a 2-port VNA, automating S-parameter measurement, visualization, and exporting across 236 points from 100 MHz to 6 GHz.

## LEADERSHIP, MEMBERSHIPS, AND HONORS

Honors College Ambassador | IEEE | SWE | Presidential Scholarship | TC Energy Scholarship | Engineering Department & Electrical Engineering Outstanding Service Awards